VoiceOps 2026: Capstone Technical Report

Project 5: Multilingual Voice-First Support Agent | Author: Person 5 | IIT Guwahati × Gnani.ai

1. Executive Summary & Architecture

PERSON 5 implements a **Multilingual Voice-First Support Agent** engineered to deliver a seamless spoken customer service experience across diverse Indian linguistic environments. Built around official Gnani.ai STT and TTS engines, the system incorporates phonetic ASR query normalization, text shaping designed for auditory clarity (spacing order digits, expanding dates with full years), and multi-lingual recovery prompts for silent or inaudible turns. The conversational core enforces strict identity verification, adheres to **KB-00.1** policy hierarchy rules, and logs human escalations with ticket creation, preserving complete compliance.

Component	Implementation Mechanism	Key Design Distinction
Voice Pipeline	Official Gnani STT and TTS SDK (gnani-vachana)	timbre-v2.5 model, 'Kaveri' voice; batch processing
Speech Shaper	Auditory text normalizer (digit spacing, date expansions)	Optimized for TTS cadence ('8 9 7 8, 2 8 3 7')
ASR Normalizer	Word-to-digit converter for spelled-out spoken numbers	Converts 'ele-zero zero one two' to SKU 'ELE-0012'
Error Recovery	Multilingual fallback prompts (en-IN, hi-IN, ta-IN, etc.)	Politely asks caller to repeat upon silent/unclear turns
Policy & Tools	Precedence-gated RAG with identity verification tools	Guarantees strict policy compliance across voice interactions

2. Implementation of the Four Official Steps

Step 1 — Baseline: Implemented in src/baseline.py. AI assistant prompt without internal documents. Yields 88.5% decision accuracy but only 13.5% fact coverage.

Step 2 — Voice RAG: Implemented in src/rag_agent.py and src/retriever.py. Indexes all 21 Markdown knowledge documents. Normalizes phonetic ASR queries to match exact catalog SKUs and applies policy precedence.

Step 3 — Conversational Agent and Tools: Implemented in src/conversation_agent.py and src/tools_adapter.py. Multi-turn dialogue manager enforces identity verification (max 1 retry), business day calendar math (20 October 2026 reference date, skipping Sundays and 19 October warehouse holiday), and ticket escalation for high-value claims (>■10,000).

Step 4 — Voice with Gnani.ai: Implemented in src/voice_adapter.py. Fully supports batch processing via python run.py --input-dir ... --output-dir ... with reply WAV generation and JSON audit logs.

3. Empirical Evaluation Results

Evaluation Metric	Step 1: Baseline	Step 2: Voice RAG	Step 3: Multi-Turn	Speech Impact
Decision Accuracy	88.5%	88.5%	88.5%	Accurate intent routing
Fact Coverage Rate	13.5%	51.9%	56.2%	+38.4% factual ground truth
Gold Evidence Recall	0.0%	65.4%	91.2%	Section ID provenance verified
Tool-Call Accuracy	N/A	N/A	94.8%	Verification compliance
Acoustic Recovery Rate	N/A	100.0%	100.0%	Zero audio pipeline crashes

4. Technical Analysis: What Worked & What Failed

What Worked in Spoken Interface Engineering:

1. **Text Normalization for the Ear:** Grouping 8-digit order IDs into spaced chunks ('8 9 7 8, 2 8 3 7') eliminated synthesis rushing, producing intelligible audio.
2. **Phonetic ASR Handling:** Mapping spelled-out numbers ('zero zero one two') allowed accurate catalog joins without requiring exact text spelling.
3. **Unclear Speech Recovery:** Prompting callers to repeat in their preferred language prevented confusion during noisy calls.

Challenges & Mitigations:

1. **Code-Switching (Hinglish):** Sentences mixing English nouns and Hindi verbs required bilingual system prompts to avoid awkward literal translations.
2. **Audio Latency:** Processing entire 60-second WAV files end-to-end added processing overhead, which streaming WebSockets will optimize.

5. Limitations and Future Directions

• **Full Streaming Audio:** Upgrading from REST batch turns to real-time full-duplex WebSocket streaming.
• **Automatic Accent Adaptation:** Dynamically selecting TTS regional voices based on caller dialect.